UCL DEPARTMENT OF MEDICAL PHYSICS AND
BIOMEDICAL ENGINEERING

Undergraduate Project Tracking Sheet

Module Code:	**MPHY0012**		
Module Title :	**Medical Physics and Biomedical Engineering Research Project**		
Coursework Title :	**Final Report**		
First Supervisor	**Henry Lancashire**	**1st supervisor's e-mail**	**h.lancashire@ucl.ac.uk**
Second Supervisor	**Prabhav Reddy**	**2nd supervisors e-mail**	**p.reddy@ucl.ac.uk**
Student's Name:	**Chuqi Wang**		
Student's e-mail address:	**zcemane@ucl.ac.uk**		

Coursework Deadline:	**17 May 2022 (ECs)**
For Office Use Only	
Date Received:	
Date Returned to Student:	

Before completing and submitting this piece of coursework please check that you are fully aware of the MPBE policy on plagiarism (See your Student Handbook).

Also note the penalties for late submission which can be found in the Academic Manual at the following link:

http://www.ucl.ac.uk/basc/current/assessment/latesubmission

Mark (%):

Please note that the mark is provisional and could be changed when the exam boards meet to moderate marks.

For Office Use Only
Reason for late submission of course work:

Table of Contents

Biofeedback with EMG Sonification and Music

First supervisor: Henry Lancashire
Second supervisor: Prabhav Reddy
Student: Chuqi Wang

Abstract

Auditory biofeedback has various applications on gaining better body control such as heart rate, muscle tension and brain function. Techniques for detecting the imperceptible physical signals and translating the data into sound are challenging. Building a prototype that completes the loop of detection and sonification of electromyogram (EMG) signal in real-time creates a chance of biofeedback application. The prototype used open source sensor and microcontroller, and an extra self-built data processing circuit. The final design of the prototype had successfully played the sonified sound with EMG signal recording in real time. The EMG feature was mapped with the frequency of the audio and a correlation between force generation and pitch changing was detectable. This study provided a starting point for developing a portable device that converts muscle signals into sound signals for the purpose of delivering precise control, and usually involuntary control to ease symptoms.

Introduction

Biofeedback using physiological data measured in real-time have a broad application. The biofeedback technique works by recording information from the body and providing visual or audio feedback to the user. The aim of biofeedback is to increase participant's attention of certain body signals, and positive feedback loop could be initiated for correction or achieving better performance. For instance, biofeedback has been used for muscle tension control (pelvic floor muscle control) and body function control (temperature, heart rate and respiratory control). The human-machine interaction introduces by biofeedback provide more approaches of understanding and using the signals of the body.

Electromyography (EMG) is a technique which measures the muscle electrical activity in response to the nervous system. It is mostly used in diagnosing neuromuscular abnormalities [1]. The usage of surface EMG (sEMG) is non-invasive, and it could be recorded for various motions. Comparing to other diagnosis and monitoring technologies, the EMG recording electronics has a compact size and easy to operate. Also, no specific environmental condition is required for recording. However, for the best quality of the recorded signals, the sensor (signal detection, noise rejection and resolution) and electrodes (size, placement, and inter-distance) are crucial. The sensor location would depend on the specific muscle anatomy [2].

sEMG studied in this research was all recorded on the forearm (extensor carpi radialis brevis), and the reference electrode was placed near the elbow.

The process of translating EMG signal as data source into non-speech audio is called sonification [3]. The information of data is aimed to be understand in the sound produced by the human auditory system. Sonification can be a way to present the information to the patient in auditory biofeedback.

The purpose of this study is to test the possibilities of parameter mapping of raw EMG signals and the significance of the sound for healthcare application. Both the EMG pattern representations and the process of human auditory nervous system translate the sound wave into biological signals affect the biofeedback outcome. Music as a well-developed representation of the auditory cognition could be effectively applied for communication the physiological information. The music can deliver some universal message. The potential of various music features and different instruments provide chances of mapping and translating the biological (nature) data [4]. A valid translation would convert information from source into audio output which could then generate the desired feedback in the patient's body. The feedback loop could train the patient to actively control their body.

EMG signal, as the data source of biofeedback has its own advantages and disadvantages. There was a little linearity between EMG and the actual mechanism information of the motion such as the stress, strain, and torque [5]. The measured EMG pattern is affected by many factors, anatomy, neuromuscular function, and different kinds of noises. However, muscle signal has its advantage to use for the real-time application in biofeedback. As the information condensed in the EMG signal would be easily missed with visual representation, the overload EMG data could have a good match with sonification (auditory translation) [6].

EMG techniques are developed in the direction of having better understanding its nature. The EMG potential is generated with the contraction of muscle fibers. The action potential arrives from motoneuron propagates forward in the muscle fiber and down to the tubules system which regulate the calcium concentration. The released calcium in sarcoplasm bind to contractile filaments. The filaments then bind with each other which cause a contraction [7].

EMG study, sonification techniques and biofeedback application constitutes this study. There are many existing theories and research in these fields that provide inspiring thoughts for the development of this project.

Here the EMG signals is not directly used to assist the muscle control and movements. One additional step, sonification is added (EMG reading digitalised for sound synthesis). Auditory nervous system contributes to the final action of participant. For EMG control of motion, there are approaches share more or less similarities. EMG features could directly control the force introduced in the prosthesis or assistive device [1]. EMG biofeedback with a visual display about the real-time muscle signals was studied to improving the grasping force in prosthesis [8]. Additionally, an instrument using analog EMG measurements was developed for assisting deep muscle relaxation [9].

Sonification with different type of data source to map with various sound parameters provide non-systematic method. A taxonomy suggested by Hermann [10] discussed the difference between sonification technique and the conditions to meet. Audification and parameter-mapping sonification was examined in the study. The ideas collected from electrocardiogram (EEG) sonification approaches [11] [12] are the most applicable for this project as EEG signal is also a biopotential.

Biofeedback includes visual and auditory biofeedback. Both types of biofeedback have a positive impact on motor control [13] [14]. EMG signal with auditory biofeedback could have a more selective representation of data as the translated sound could be altered with different extracted features. It would be easier to carry out the given task without paying the additional attention on the visual display. Also, auditory biofeedback could be applied on blind participant.

This study used two open source electronics, MyoWare sensor (SparkFun AT-04-001) and Arduino (UNO R3). The most useful EMG frequency is around 50 Hz to 500 Hz whereas the majority of noise have frequency below 50 Hz (mains interference at 50 Hz). The prototype is aimed to capture the useful signal and minimise the noise before processing the data. The EMG data was extracted in incremental window and mapped with sound parameters. EMG features like mean absolute value (MAV), gradient and root mean square (RMS) value were calculated to map with sound amplitude and frequency.

The aim of the research was to develop a feasible prototype that convert EMG signals into musical sound. The prototype should have a possible auditory biofeedback application.
In order to have a successful device, there are a few objectives for the divided steps of building the prototype (Fig. 1).

1. Collecting characteristic and informative EMG signals by recording different actions and grasps.
2. Using microcontroller and MATLAB programming to extract features from recorded EMG signals.
3. Developing a significant algorithm for the EMG features and (musical) sound mapping by researching and examining different approaches.
4. Testing any change of controlling muscles with listening to the audio synthesise by the built prototype.

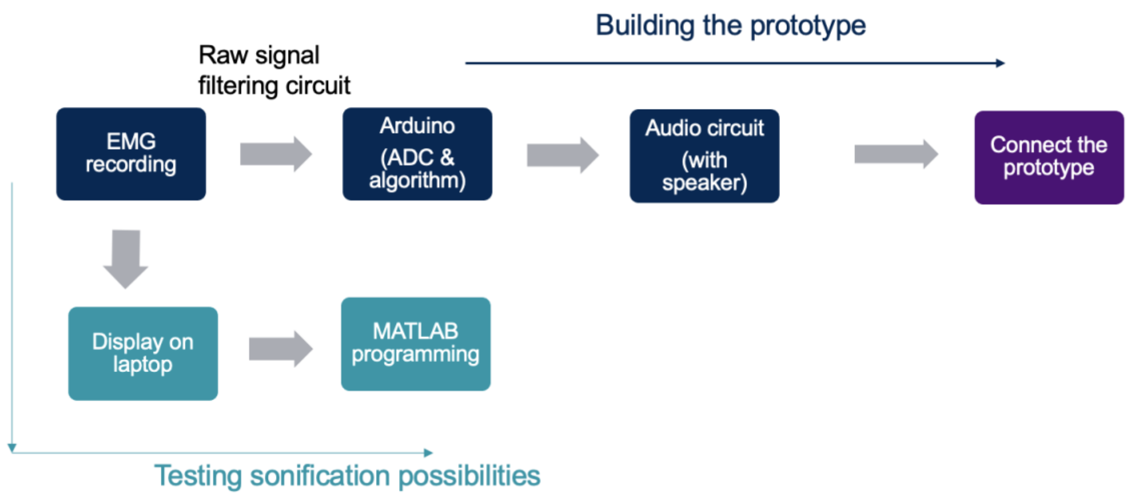


Fig. 1. The process flow diagram of building the prototype

Methods

A. Data collection for initial analysis

1) Recording setup

Main components include: EMG sensor, electrodes, Arduino, laptop

The MyoWare sensor (SparkFun AT-04-001) has two output modes, EMG Envelope (SIG) and Raw EMG (RAW) [15]. The Envelope output is a pre-processed signal (within the sensor) that design to work with a microcontroller like Arduino. The rectified and integrated signal worked with the default Arduino communication serial rate for sampling. RAW has an offset of Vs/2 (Vs is the supplied voltage to the sensor) and requires a much higher sample displaying rate for storing the spike-like signal.

Three electrodes (BIOPAC EL500, Ag/AgCl disposable snap electrodes) were placed on the participant forearm, two recording electrode on the extensor carpi radialis brevis were separated by 3 cm (the distance between the sensor snaps). The mid and End Muscle Electrode Snaps of the sensor was clipped onto the recording electrodes and the black reference snap was attached to the electrode near the elbow.

RAW and SIG were recorded via (Arduino) analog input pin, A0 and A1 respectively. During the recording, the sensor was powered by two cell batteries (3 V each). The GND pins on sensor and Arduino was connected. Two separate Arduino scripts were written for Raw and Envelope recordings as the required data displayed rate were different (appendix). There were two approaches of Raw output which would be mentioned later. Arduino was communicated with the laptop via the serial.

2) Experimental procedure

A few different wrist movements were recorded for testing the sonification mapping in Matlab. The start and end time of each recording was tracked and matched to the timestamp of the Arduino monitor. Recordings included:

- i Pulsed Extension: raise wrist to the maximum extension angle within 1 second and relax in 1 second (gently with a uniform speed). This was repeated for 5 times and a clock was used to track the whole process.
- ii Pulsed Extension with resistance: same as above, with the other hand pushes on the back of the moving hand, resistance was applied during the raising of hand
- iii Holding the extension: quickly raise the wrist to the maximum angle and hold this position for 5 seconds.
- iv Radial deviation: bend wrist towards the thumb in 1 second and return to the normal position in 1 second.
- v Ulnar deviation: similar to radial deviation, with wrist bending outwards (towards little finger).

Both Raw and Envelope signals were collected for the above wrist movements. All Envelope recordings were done at one time (keep the electrodes on between recordings) and Raw signals were recorded at a latter day with same setup and different Arduino scripts.

B. Raw EMG signal

1) Raw recording

   i The sensor was battery powered and with the same recording configuration as mentioned. In the Arduino sketch, the baud rate was changed from the default, 9600 bits per second (bps) to 115200 bps. The readings were converted back into float voltage value before printed to the serial port. The values were then printed with Serial Monitor tool and stored with timestamp. This refers to the “Raw 1” script (in appendix).

   ii The second approach (“Raw 2” script) of acquiring a higher displayed sampling rate was to group the readings. 500 data points were stored in an array and sent together through the serial once the array reached this size. To ensure a stable switch between recording loop and sending loop, there was a 1 millisecond “delay” between two loops. The default baud rate (9600 bps) was used (appendix).

2) Raw signal processing

An external circuit was designed for Raw output of the sensor to filter out the noises and amplify the signal. The processed raw signal will have less misinterpretation in the algorithm.

The operational amplifier (op amp) was found in RS Components. The circuit need an op amp single supply with an operating voltage above 5 V (voltage regulated by Arduino). Also, the op amp should have a rail-to-rail output that enable the mapping of output signal to the whole voltage range of the supply). Features like small bias current means less error bought to the signals by the op amp. Considering all of this factor and the price, MCP606 was chosen for this designed circuit (Table 1).

type of op amps	supply voltage	rail-to-rail output	maximum bias current	typical slew rate
MCP606	2.7 V to 6.0 V	yes	80 pA	0.08 $V/\mu s$
TLE2021CP	5 V to 28 V	not specified	not specified	0.5 $V/\mu s$
OPA350PA	2.5 V to 5 V	yes	not specified	22 $V/\mu s$
MACP6021	2.5 V to 5.5 V	yes	150 pA	7.0 $V/\mu s$

Table. 1. Information of the op amps that considered for this circuit

The circuit contained a high-pass filter with unity gain. A second order Sallen-key configuration was applied (Fig.2). The calculated cut-off frequency (with sine wave) from transfer function was around 92 Hz.

$$\mathrm{H(s)} = \frac{s^2}{s^2 + s\left(\frac{1}{R_2 C_1} + \frac{1}{R_2 C_2}\right) + \frac{1}{R_1 C_1 R_2 C_2}}$$

In the transfer function, s is complex angular frequency, the resistor and capacitor values are stated in diagram and table 2. The estimated bandwidth is 30 Hz.

A non-inverting amplifier circuit contained a 1 $k\Omega$ and 10 $k\Omega$ resistors to introduce a gain of 11:

$$\frac{\mathrm{V_{out}}}{\mathrm{V_{in}}} = 1 + \frac{\mathrm{R_{out}}}{\mathrm{R_{in}}}$$

High pass filter	amplifier	Voltage follower
$R_1 = 10\ k\Omega$	$R_3 = 1\ k\Omega$	$R_5 = 100\ k\Omega$
$R_2 = 30\ k\Omega$	$R_4 = 10\ k\Omega$	$R_6 = 100\ k\Omega$
$C_1 = 100\ nF$	/	$C_3 = 100\ nF$
$C_2 = 100\ nF$	/	/

Table. 2. Components (resistors and capacitors) used in the circuit

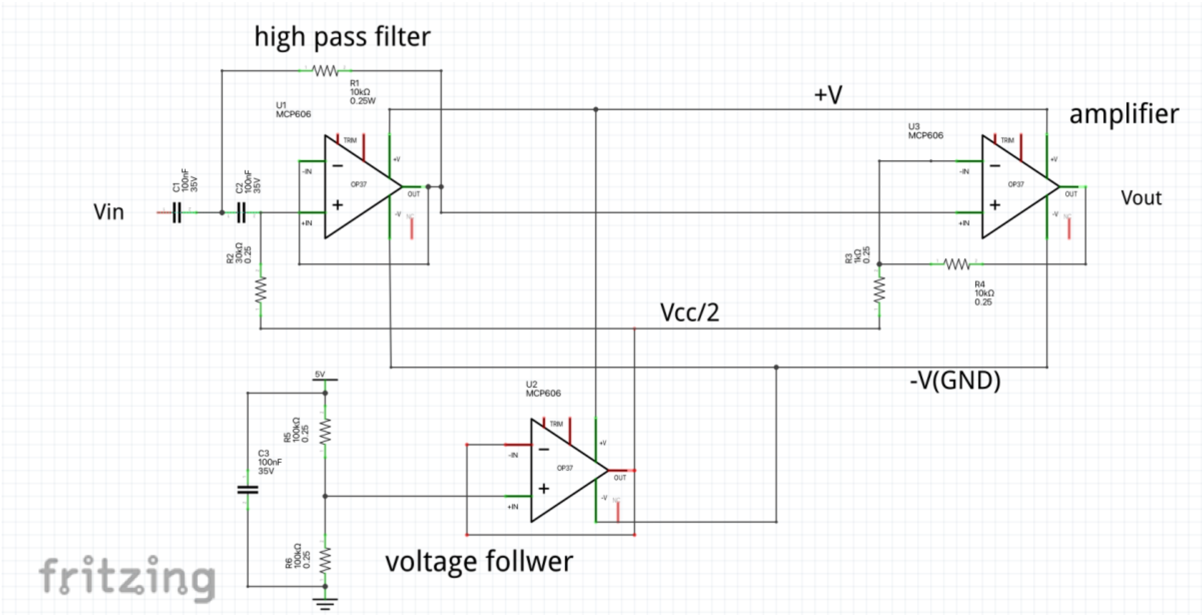


Fig. 2. Schematic diagram of the external circuit, with high pass filter (left), amplifier (right) and voltage follower (bottom)

The amplification was carried out after the filtering stage. The original circuit with filtering and amplifying stage in one Sallen-key filter design did not provide the expected output signals. The output amplitude was peaked at 60 Hz and no amplification was provided above this frequency. In the raw signal circuit, a stable centred voltage (+Vs/2) was set as the virtual ground of op amps using a voltage follower (Fig. 2).

The whole circuit was firstly constructed on a breadboard. Response of the circuit was tested with oscilloscope and signal generator. The power supply was set to 5 V. Input sine wave with small peak-to-peak amplitude (in mV) and 2.5 V offset was generated to simulate the signal processed in the prototype. The change in peak-to-peak signal voltage and phase shift were examined across 10 Hz to 1000 Hz. Then the tested design was soldered onto a veroboard. The Raw output was connected to the external circuit and the processed Raw entered the Arduino pin A0. The SIG (envelope pin) passed through the Veroboard without entering the design circuit (Fig. 3). The prototype was supplied by Arduino and the voltage was regulated at 5 V as suggested in MyoWare manual [15].

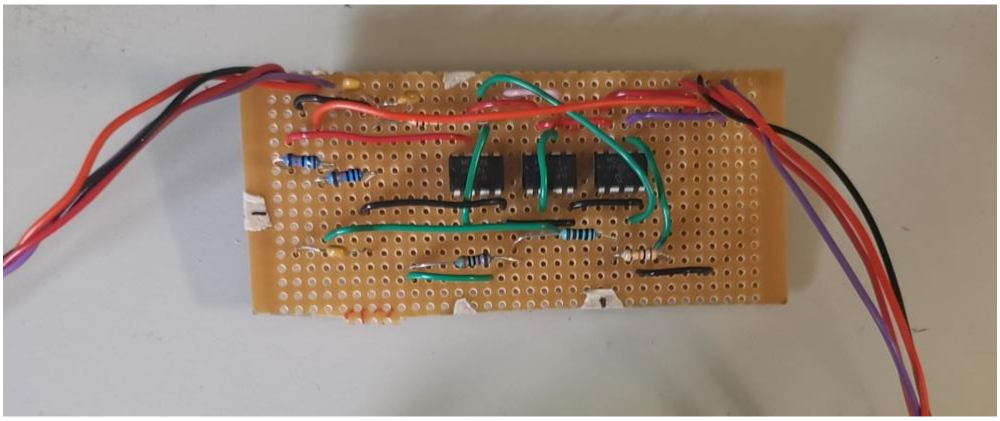


Fig. 3. A picture of the external circuit for raw signal processing, colour code of wires: red - positive power supply; black - ground; orange – envelope signal; purple – raw signal

C. Sonification methods

1) Matlab programming using recorded EMG data

The data were gathered from the participant forearm muscles. The data was collected the raw recording mentioned above were stored in a laptop. Full script was included in appendix, called “Matlab sonification”. The data were gathered from the participant forearm muscles. The result from both Raw recording scripts mentioned above were stored in a laptop. The EMG was firstly plotted with the timestamp. The coding was divided into few sections, the steps included:

- i Raw data rectification. To recentre the raw signal back to zero (remove the offset).
- ii 50 ms time window creation. A reference of time increments below 125 ms in real-time application was given, the length of time window could be varied to acquire the best attention and control of participant later.
- iii Parameters extraction loop. Calculate parameters in one time window within one loop
- iv Sampling sine wave (sound wave) creation. A high sample rate (40000 Hz) was chosen for sound synthesis.
- v Parameter mapping. Transform EMG into sound signal.
- vi Write the audio data into file.

Mean absolute gradient and MAV (Eq.1) of the EMG signal were used for mapping with sound parameters for sonification in my testing stage.

$$MAV = \frac{\sum_{i=1}^{n} |V_s|}{n} \quad (1)$$

where V_s is the sample voltage, n is number of samples in a 50 ms window.

$$g_i = V_{i+1} - V_i \quad (2)$$

where g_i is the gradient at sample i.

$$mean\ absolute\ gradient = \frac{\sum_{i=1}^{n} |g_i|}{n} \quad (3)$$

The raw data has an offset at Vs/2 which determined by the sensor power supply. In the case that the sensor is powered by battery, the offset would vary with battery consumption. A general approach here was to determine the offset by average all samples in one recording. Data was zeroed by subtracting the mean from every sample point. The gradient was simply calculated by shifting the data array index circularly by one. The difference between the adjacent values was counted as the gradient (Eq. 2). As the cancellation between positive and negative gradients provided no significant information about the muscle and movement, the mean absolute gradient was applied here (Eq. 3).

For the time window division, the sample size within 50 ms might vary depending on the overall sampling rate in each trail (for example, sample reading speed, calculation speed and serial communication rate). It was assumed that each time window has same amount of data. The number of windows was calculated using the total recording time referred by the timestamp of data. The closest (integer) number of samples was grouped for window calculation.

The parameter extraction loop ran the calculations for all windows and each parameter was stored in an empty array. The extracted factors were plotted in Fig. 4. These parameters were then normalised. The sound sine wave with sampling frequency (f_s) at 40000 Hz created (Eq. 4), with a same duration, 50 ms.

$$y_1 = \sin(t_s\, 2\pi) \quad (4)$$

where t_s is a time vector from 0 to 50 ms, with interval of $1/f_s$

$$y_2 = ax_1 \sin(t_s\, 2\pi\, bx_2) \quad (5)$$

x_1 and x_2 are normalised variables, MAV and mean absolute value. With coefficients a and b which indicate the weight of mapping

MAV was used to control the frequency (pitch) of the sound and the mean absolute gradient varied the amplitude (volume) of each tone (Eq. 5). As the human sensitivity to sound was in a logarithmic scale (in dB), a logarithm change in frequency and amplitude with EMG parameters was also tested.

$$y_3 = x_1 \sin(t_s\, 2\pi\, b * log_2(x_2))$$

The audio file was written by “audiowrite” in Matlab. In addition, the generated sound was compared between different mapping approaches.

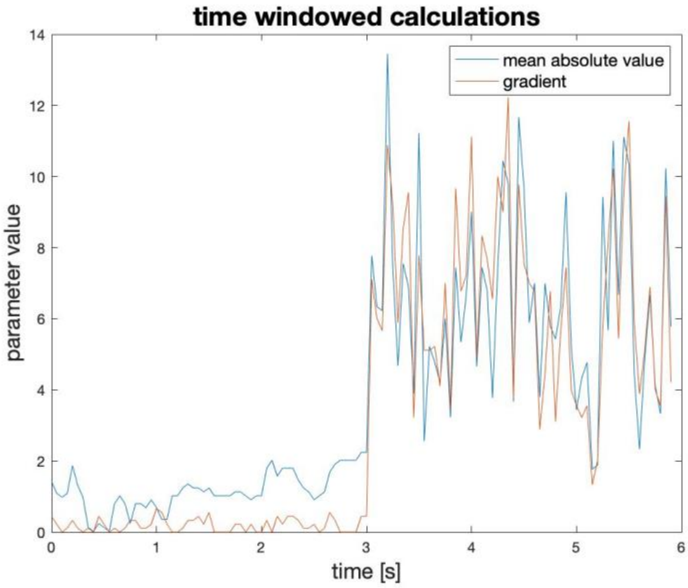


Fig. 4. Extracted raw EMG parameters in time windows for a total recording length of 6s

2) Arduino script ran in the prototype

The extraction and mapping method was similar to the Matlab coding (Fig. 5). A few modifications were included to either speed up the running time of commands or correct the algorithm working with a different language here. Assigned variables were cleared at the end of loop to avoid the building up from previous looping. The data type of variables was assigned before the main coding. For value with known size, appropriate data type could occupy less storage to speed up the loop. As the prototype was powered by a regulated power source, the offset of Raw mode kept stable at 2.5 V, which with a digital representation of 512 (1024/2). This was subtracted from all raw readings before working with the data. There were three sections of coding to separate the recording from the time-consuming calculations. The Arduino recorded readings into an array until the size reached 50. Then it entered the loop for parameter extraction. One tone was generated as a single window loop was completed. Tone function generated a square wave at the specified digital output pin with frequency varied with MAV. A multiple factor (reference factor) was used for frequency mapping adjustment. It was set to 100 in the script, and it could be adapted for different recordings. The "tone" was set with no duration, so the previous pitch played continuously until the next tone was initiated. To prevent the summation of values from the previous window in the new window, the window variables was reset to zero after playing the tone. Also, the EMG recording array was emptied before starting the new recording ("Arduino sonification", in appendix). Furthermore, the serial monitor was used as debug tool to track the breaking point within loop during the test stage.

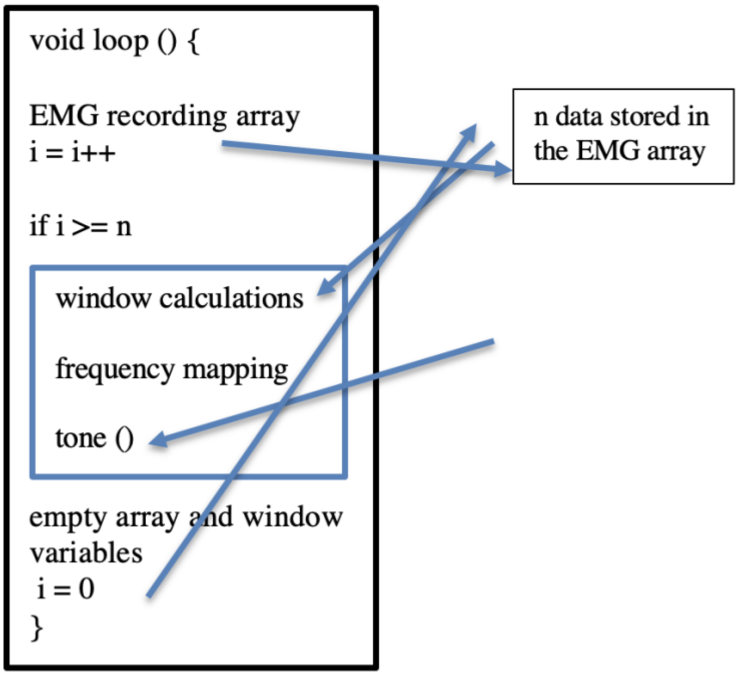


Fig. 5. The basic algorithm running in the prototype

D. Hardware (prototype)

The prototype components involved the sensor (MyoWare), the microcontroller (Arduino), the self-built raw signal filtering circuit and audio circuit. The external filtering circuit was added between sensor and Arduino. The accessories of MyoWare sensor included a cable with 3 electrode pad connectors, it had a very poor connection when testing in forearm muscle recording. The noise level was high so that the useful signal could not be indicated. Wires with appropriate length (around 35 cm) were soldered to pin header connectors and twisted together to strength the connection (Fig. 6). The pin connectors fitted into the sensor pins and the other end of the wires were soldered onto the veroboard. Similar on the other end of the board, connectors were used for power and analog input pins on Arduino. This provided a better connection and a movable range with the sensor. The code ran in the Arduino extracted features from sensor readings and apply them to produce the audio output ("tone" function). A continuous, pitch-changing sound was played by the speaker. The audio circuit tested included a speaker (ABS-222-150-RC, 8Ω), a 100 Ω resistor and a variable resistor (Fig. 7). The resistor was chosen with the requirement that, this 1W speaker in a circuit with maximum voltage of 5 V, required the current to be restricted below 0.2 A. Variable resistor was adjusted to vary the sound volume.

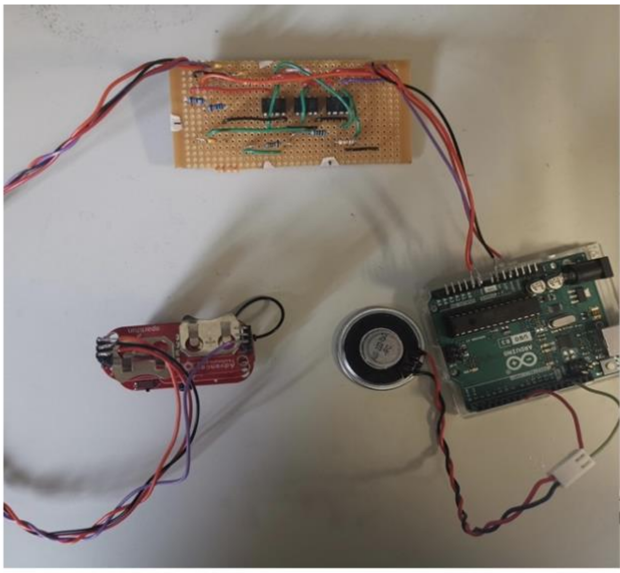


Fig. 6. A picture of the prototype

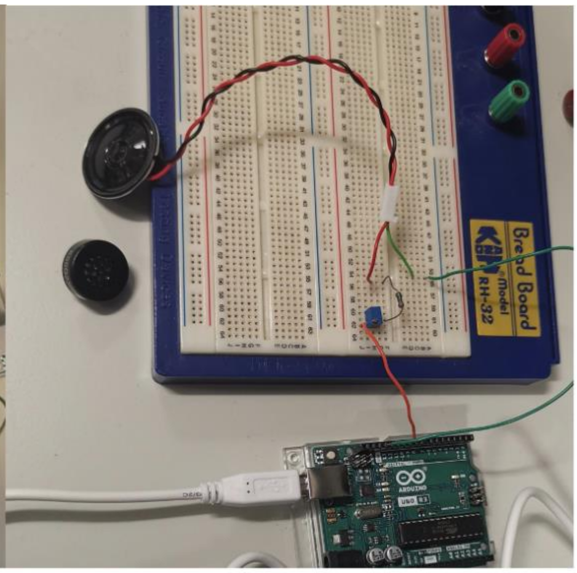
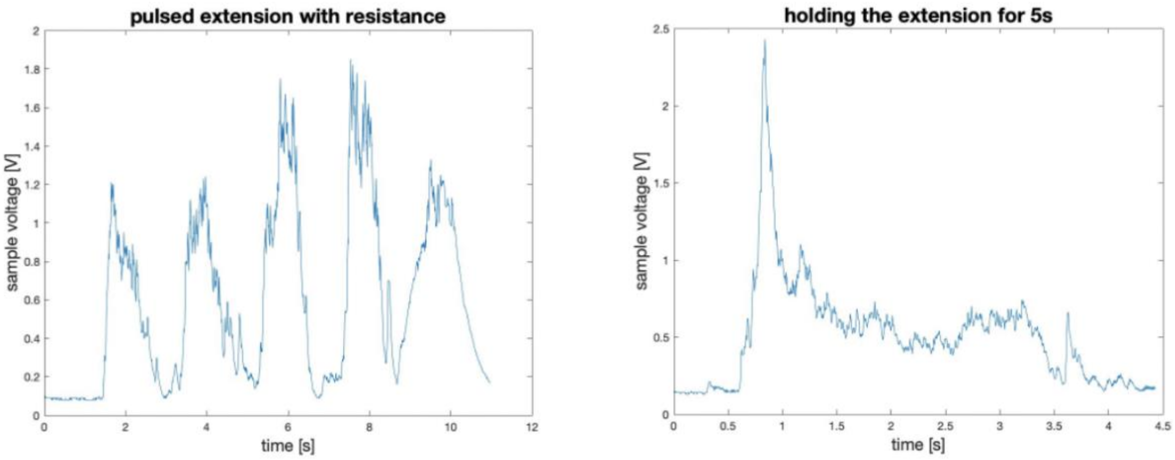


Fig. 7. The audio circuit design

Results

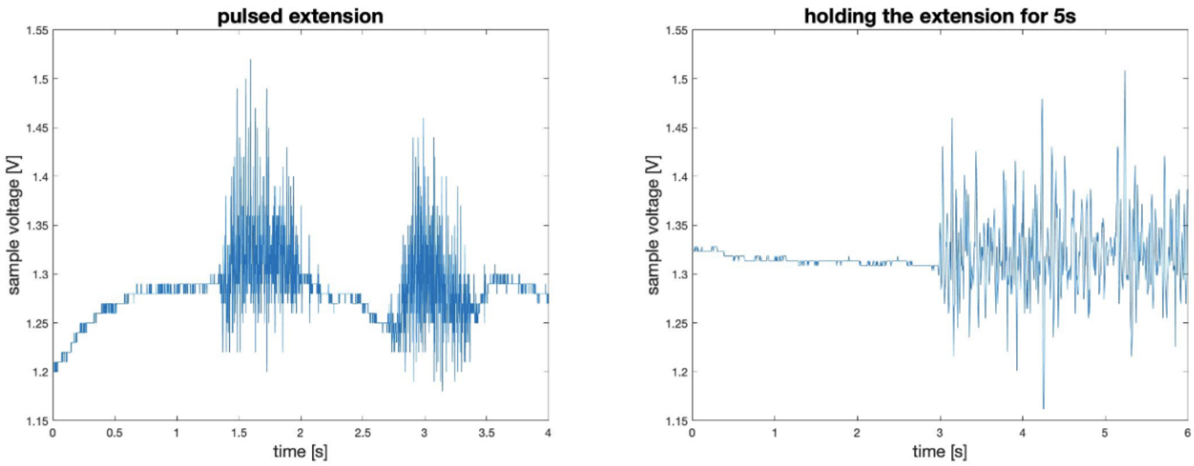
A. EMG recording

The envelope recordings with different wrist movement are shown in Fig. 8. With the serial monitor tool (9600 baud), there was around 180 samples collected per second (180 Hz). The raw EMG was recorded with two different Arduino scripts as mentioned in methods (Fig. 9) Data was copied from the “Serial Monitor” tool after the recording based on the timestamp marked. The digital readings were converted back to the corresponding voltage and time resolution in the plot was corresponded to the displayed sampling frequency.



a) b)

Fig. 8. Envelope recording of a) pulsed wrist extension (with resistance); b) 5s wrist holding at its maximum angle



a) b)

Fig. 9.a) method 1, high baud rate (115200); b) method 2, storing and sending loop

B. Raw signal pre-processing

The original circuit design in the prototype was a second order Sallen-key filter with a characteristic gain: a non-inverting filtering and an inverting amplifying stage configured together with one op amp (Fig. 10). However, result indicated a maximum output amplitude around 60 Hz (1.8 gain), the signal then reduced until reached close to 120% of the input.

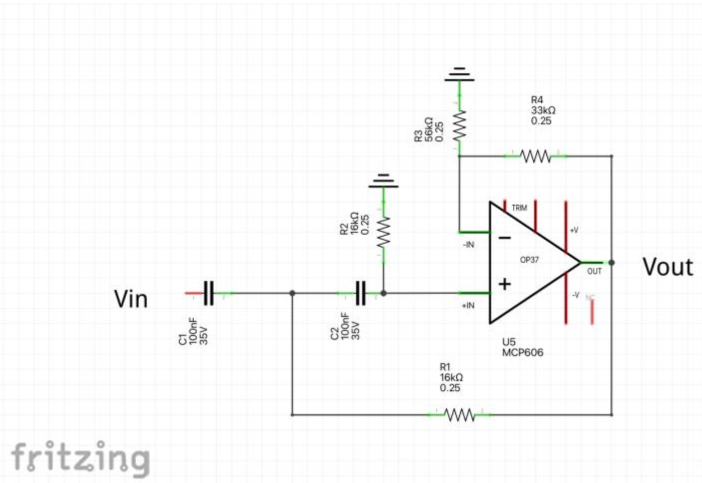
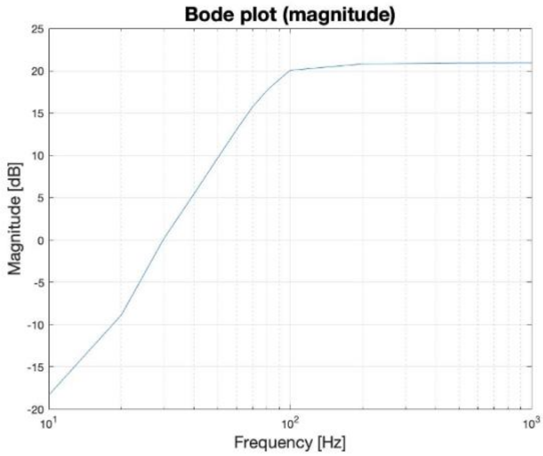
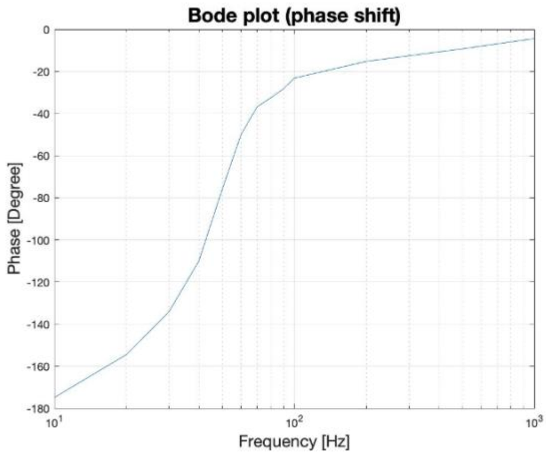


Fig. 10 Sallen-key filter with non-unity gain (amplifying with the ratio of $\frac{R_3+R_4}{R_3}$)

Therefore, the two stages were separated. The overall outcome of the external circuit was tested with a signal generator. Bode plot (Fig. 11) was shown below. By testing across 10 Hz to 1000 Hz, 30 Hz signal had an overall gain of 1 and the 11-fold of magnitude was reached at close to 200 Hz. The phase shift angle decreased as the frequency increased. A 180° phase difference was measured at small frequency (10 Hz) and the input and output signals were in phase at higher frequency.



a)



b)

Fig. 11. Bode plot of a) magnitude; b) phase shift over the tested frequency range (10 Hz to 1000 Hz)

The raw recording with external circuit were shown below (Fig. 12). There was a much stable baseline at rest (in the first second of recording). The pattern was well centred at the offset, 2.5V.

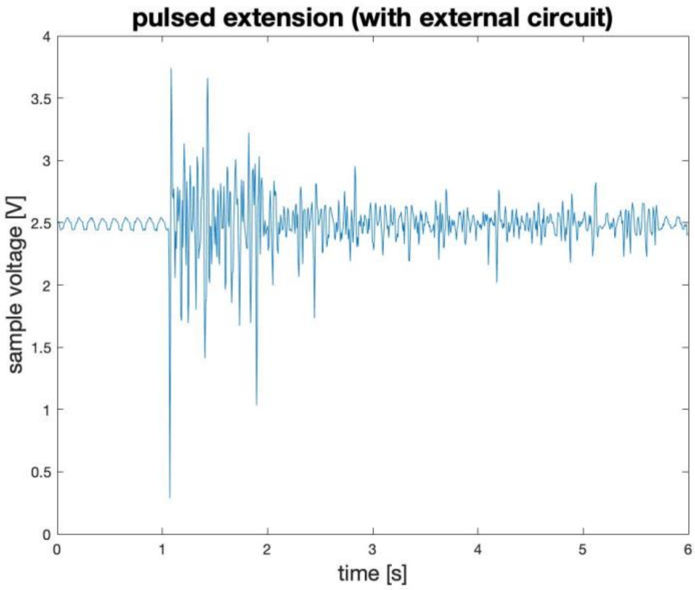


Fig. 12. The pre-processed raw data plot with pulsed extension

The filtering and amplifying were also demonstrated in Matlab (Fig. 13), raw signal recording without external circuit was used. A high pass Butterworth filter was added, and sample amplitude was amplified with a factor of 10.

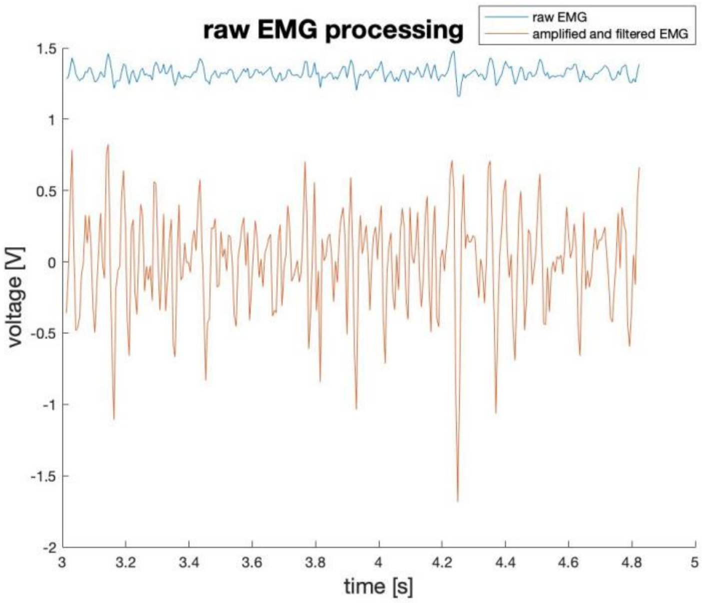


Fig. 13. A section of raw EMG and the amplified and filtered EMG plot on the same graph

C. Audio output

Audio circuit in the prototype was operated with the "tone" function in the Arduino. The sound volume control was not inbuilt in this function. The only audio parameter adjusted in the prototype was the frequency. The output sound was a continuous wave with frequency changed based on each EMG window. The chosen reference factor of frequency was 100 and a linear relationship between MAV of raw EMG and tone frequency was assigned and tested successfully with the prototype in real-time. The sound played with sensor turned on but no connection to electrodes was a constant, non-changing audio. The pitch change with the level of exerted force (which controlled actively by the participant) when attached the sensor on muscles. A short increase in pitch was observable when the force increased.

In the Matlab algorithm for sound synthesis, the audio output had more variations as both pitch and volume were mapped. The audio data was plotted (Fig. 14). The change in volume would sometimes disturb the perception of pitch. Also, the artificial audio was not very musical or similar to any nature sound. Considering the short time window of each tone, it was rather difficult for the participant to extract information from the audio at the first time.

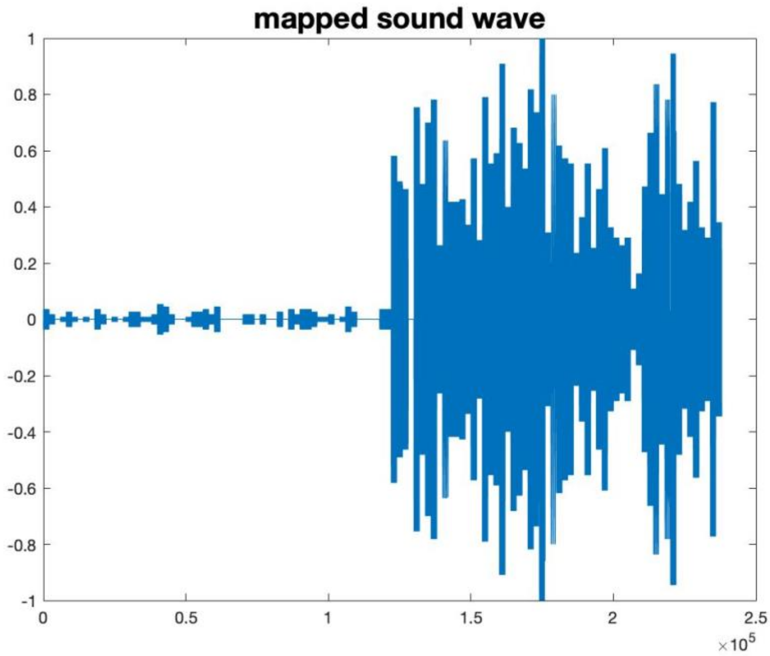


Fig. 14. The audio data plot in Matlab

Discussion

This study has successfully constructed a portable prototype which sonifies raw EMG signal into audio in real time. By using open source EMG sensor and microcontroller to build up the prototype, this device could be easily replicated and modified. Although the biofeedback possibilities have not been demonstrated on any group of participants, the audio produced from the separate programmed Matlab and the prototype were in a sensible frequency range. A linear correlation between the dynamic of the muscle signal amplitude and the sound pitch change was observable.

A 50 ms time window was chosen for data recording in Matlab sonification algorithm. It was chosen with the consideration of the significance of the extracted features. A large time window would smooth the EMG pattern. A reference time window of 125 ms were commonly used in real-time EMG pattern recognition [1] [16].

The balance between real time feedback and the best response time for patient to extract information was important. The delay time for the participant to navigate and process the information involved in sound output are crucial for the outcome of biofeedback. However, as the mapped sound window had same length as EMG feature extraction window, the information carried in EMG and received by participant would both affect by the choice of window length.

In the tested the prototype trial, instead of group the data in time window, a fixed size of EMG array was chosen. The algorithm time was an unknown. Time taken for a single loop estimated with the calculation and command speed table. The way to decrease the time consuming include select the right data type for variables and simplify the calculation. The saved space could be used to extract more EMG features and map sound with multiple parameters.

The algorithm in the Arduino could be improved by using the inbuilt timer in Arduino. The computational speed would increase as the timer allow the loop running and only interrupt at the precise chosen time intervals.

The built device is in a compact size, the configuration of the components could be improved to become neater and more convenient to wear. Also, a better arrangement could reduce the noise. Apart from the intrinsic noise which are difficult to reduce and eliminate, one of the major noises that depend on the setup was the motion artefacts. The participant recording limb should be kept at a stable position to avoid any irrelative activation. Moreover, the connections between sensor and recorded muscle would introduce noise, both the electrodes and cables attached. The MyoWare sensor used was manufactured for recording with sensor directly place on the skin, the cable with extended electrode connectors had bad performance in recording. The signal was indistinguishable from the noise caused by the swing cable with a big amplitude. The solution is to use the built-in electrode snaps on sensor with a braided wire for sensor pin connections. The impact of motion artefact was reduced significantly as the envelope signal and raw recording were feasible in this configuration. The electrode-electrolyte interface performance would also affect the recording quality. Hence, new electrodes were replaced after a few uses. The sensor was loosely connected by clipping on the snaps to the electrode. After arranging the prototype components together into one ensemble, the device could contain a fasten band to ensure that the sensor connected firmly on the skin.

The external raw circuit was designed to reduce the noise level and amplify the useful signal to fill the full range of the analog value of Arduino ADC (up to 5V). Noise source considered in this study include the mains interference, at 50 Hz, as the most common noises in electronics equipment. Other common noise such as motion noise (around 10 Hz) and the instable nature of the EMG signal [17] with frequency mainly below 20 Hz. As the useful EMG frequency range lies above the frequency range of the most noise, a high pass filter is designed for noise attenuation. The choice of amplification considered the raw spike's amplitude recorded in the study. General raw signal amplitudes were in a range of 0.3 V and with an offset at Vs/2. It is reasonable to have an approximately 10-fold amplified signal to occupy the 5 V range of Arduino ADC. The resolution of the ADC (10 bits map with 5 V) is 4.9 mV. Hence, the loss in resolution due to the signal digitalisation could be compensated by amplifying the analog signal before the conversion. The precision of measurement would be more likely to limited by the intrinsic sensor properties itself.

The observed cut-off frequency of the filter was close to 100 Hz and the signal at 50 Hz was attenuated by 70% (Table 3). The noise residual would be amplified. It was also relying on the sensor itself and common rejection ratio (CMRR) of the op amps for noise removal.

Frequency (Hz)	Input Amplitude (mV)	Output Amplitude (mV)
50	106	32
60	106	40
70	108	56
80	108	74
90	108	80
100	107	96

Table. 3. The measured output signal with the high-pass filter (input set at 100 mV, recorded the real signal measured with oscilloscope

The external circuit design has no specified processing procedure for certain EMG signal pattern. This matches with the design purpose of the prototype, to attempt a general sonification of EMG signal. This direction of initial work is aiming to provide more possibilities in future development and healthcare application.

There are many possible improvements in each stage that could build towards a better prototype for the use of biofeedback application. Including the discussion of the results from previous studies and basis in relevant fields, potential direction of future work would be indicated.

To collect characteristic and informative EMG, the relationship between muscle activities and actual recording signals is necessary to be explored. The EMG pattern would be affected by the muscle anatomy. The cross talk before reaching the recording sites, distortion in electrical equipment and other biological noises like electrocardiographical signal (ECG signal) will construct to the overall EMG pattern [17]. All of these noises and mixture of action potentials from muscle fibres would lead to a misinterpretation of recording signals in some extend. The quality of EMG detection that required for biofeedback purpose would be vary case to case. The most direct way to analysis whether the EMG recording of a study is necessary to be improved is to test the biofeedback outcome. A minimal achievement on the purpose (such as increased force exertion) would illustrate the right information is recorded. Or one step backwards, instead of measuring the motor function improvement, investigate the information stored in sonified sound.

In this study, there was a lack of general EMG feature analysis before the sonification. The complexity of the EMG signal requires various techniques in detection, processing, and classification stages [18]. There are a lot of research on the understanding EMG signal in the applications introduced. For instance, a “double-threshold detection” method used for event-based detection was discussed by Bonato’s group [19]. The chosen threshold (for muscle activation) could be modified and adopt in the practice carried by each participant with various signal patterns in different application.

Comparing to the approach here. All sample points were included into windows (where one window counted as one “event”) and produce a continuous sound once the sensor is turned on it would be a great idea to emphasise the actual event window by adding parameters control the sound wave or generating a characteristic sound (such as a strong beat) between different type of motion that could be characterised. However, there would be a concern of the computational time for real-time application in this prototype. It would be useful to add a testing phase, the relevant threshold values could be searched or calculated of specific task.

Then, putting the value into the Arduino script and comparing with sensor readings for the control of one sonification mapping parameter.

The EMG parameters was directly extracted in the time domain in this study, however, wavelet transform with mapping of time and frequency domains is a common decomposition method for EMG signal. Wavelet transform could provide the power factor of signal. The power feature could be applied in the detect the fatigue of muscle [20]. Furthermore, models such as autoregressive model can deal with the inherent instability of EMG, and artificial neural networks could train for each individual patient and recognise signals in real-time [18]. In popular EMG applications, the prosthetic control also uses real-time myoelectrical signal as source. The hardware implementation is updated to evolvable hardware chip (EHW) in some studies to enable more complex control (motion with higher degree of freedom) of the prosthesis [18]. Such models significantly increase the computational and learning ability of the data. A potential improvement in EMG signal recording of this study is to learn from those matured models and improve the hardware. Overall, keeping the portable size and battery powered feature of the prototype is prior, at the same time, more understanding and interpreting of EMG source would improve on the quality of sound synthesis and ultimately, the precise muscle control in biofeedback.

At the start of sonification stage, the raw EMG was played in the Matlab by directly using the sampled EMG as audio data (audification) [3]. The output was a noise-like sound. A window with high sound intensity represented an active period of muscle signal. The translated sound was shown with no meaningful feature about the specific activity. This suggested that a better mapping algorithm for the audio output is necessary.

EMG as the sonification source is not very common and lack of systematic approaches [6]. The possible reasons could be the nonlinearity of the measured pattern and the actual motor function. Also, the evaluation techniques for EMG signal have a high computational demand as discussed. However, the fact that EMG provides a detailed information of neuromuscular activities. EMG pattern comprises with a dynamic muscle signal which is generated by the temporal structure change of muscle. The formation of EMG pattern shares with some similarities with the synthesis of a piece of artificial sound (temporal amplitude and frequency change). In addition, using the auditory perception to listen to the translated physiological data has a faster event detection rate than using visual feedback. Physiological event such as muscle activation is less detectable in the visual display [6]. The parameter mapping for translating data would have a big impact on the participant's understanding of the information. The combination of EMG and sound features choose for sonification and the human responses to the created stimulus (sound) contribute to the body feedback.

The auditory nervous system deal with the background sound (receptive listening) and look for the one that human pays extra attention on to determine some sound properties (mining) [21]. Here in the case of meeting the need of presenting the physiological information, the participant should have a sensitive perception of the translated sound. The general mechanism of the sound energy transduction includes a multiple steps in the outer, middle, and inner ear [3]. The sound is converted into biological signal when reach the nervous system. Both frequency and temporal resolution are kept high in the transduction process and the tuning at basilar membrane (with changing stiffness along the membrane location). The basilar membrane detects the signal with the pressure change and code the frequency with the resonate basilar location (tonotopic representation) [3]. For pitch below 4 kHz, the accuracy of frequency coding is high as the induced action potentials will be in phase with the sound stimulus. For the sound intensity detection [3], the sensitivity (the smallest distinguishable change) is described with a logarithmic scale (dB). The sensitivity is mostly proportional to the total intensity (Weber's law) [3].

While the threshold sensitivity would vary with frequency (peak at around 4 kHz) and duration (maxima at 100 ms to 200 ms) [3], the sound falls in these ranges would be easier for people to recognise and study. The information about the sound transduction process and psychoacoustics provides a good reference for sound frequency and amplitude range choosing for the sonification in the prototype. The aim of constraining the pitch, volume and duration of sound is to optimise the perception of data delivered in sound.

Different from sensing the natural sound, music provides more emotional influence [22], this then leave with more possibilities of acquiring physiological information from the recognition of music features like timbre, melody, harmony, and rhythm. The human understanding of music is based on the evolutionary process and study [3].

The prototype was lack of consideration in the sonification method, which corresponded to the quality of the sonified sound. In the Arduino, "tone" function only provides frequency and duration variations, and the sound was not very “musical”. The “tone” produces a square wave audio signal wave. The tone duration was not pre-set, and it is depended on the processing time of EMG feature extraction. This ensured a real-time approach, and the listener could focus on the pitch change without the disruption of other variables. However, less information could be coded to this synthetic sound. This could be improved by assigning more EMG features to the single variable, tone frequency. By testing on different weight of source parameters with various combination, a better sonification solution could be introduced. Additionally, the less pleasant square wave could be adjusted with the pulse width modulation (PWM). PWM provides control of the analog sound output with digital parameter [23].

Another limitation of quality of sound was the audio circuit itself. The tested final design of prototype only consisted with a resistor (to limit current and therefore the volume) and a speaker. In the testing stage, a variable resistor was added to the circuit (Fig. 7). By altering the resistance with a screwdriver, the audio volume was adjusted. This manual adjustment only enabled the modulation between different recordings, it was not correlated with the physiological recording data. A transistor could be added to control the sound volume played by the speaker.

The sonified sound in Matlab has a constant duration as the EMG time window was fixed. To preserve a same length audio, each pitch was played in the duration of the time window. The sound parameters mapped were the frequency and amplitude which are the most dominate features use in sonification. More possibilities could be tested in the Matlab sound synthesis as more than one parameter could map to a sound parameter. The frequency generated could be sitting in the desirable range. The normalised parameter values (MAV, mean absolute gradient and MRS) were set between 0 to 1. The reference coefficient of the EMG features could then be used to choose the frequency range and another constant could be added as offset.

$$mapping\ frequency\ range = (nomalised\ value * A) + C$$

Any impact of different musical sound translation approach could be tested with a specific biofeedback as future work of using this prototype. For instance, developing a test on the wrist motion control (as wrist motion EMG pattern was recorded and studied here). By sonifying the EMG pattern for wrist extension and flexion into audio. Investigating whether natural and half-way extension and flexion positions could be quicker achieved by the participant. No visual participation is allowed in this test. This example experiment could be conducted after modifying the prototype.

The prototype using EMG sonification for biofeedback application has no systematic approach. In the field of biofeedback, the previous studies either using physiological signal input, converting the physical information into audio (auditory biofeedback) or aiming for muscle control provided useful guidance for this study.

There are published study on biofeedback using EMG signal. The applications including muscle strengthening and treatment of neuromuscular disease. It was proved that in certain situations (for example, quadriceps femoris muscle control with meniscectomy, muscular tension reduction and gait function), the inclusion of EMG biofeedback would deliver better outcomes than with only electrical stimulation [25]. However, the existing of positive impact by adding biofeedback is arguable and should be discussed based on each case study. The data collect from the randomised controlled trials (RCT) [25] is meaningful to proof the significance. For the biofeedback possibilities here with the prototype, the participants should be given with same tasks of motion and the performances could be compared in a RCT design.

Additionally, other similar physiological data such as electroencephalogram (EEG) signal was well studied for auditory biofeedback. There are some similar ideas and appliable techniques that could be referred in EMG. As the sampling frequency of biological signal recording was much smaller than the sound sampling rate, it is hard to find the correlation between two types of signals directly [11], that is why audification of either EMG or EEG are less meaningful than sonification which compute the source input to a perceptible range of sound. The benefits of EEG sonification is that there are many clear features in the signal which could be mapped with musical features. Generally, a main feature for investigation is its rhythmicity, there are non-random sequences waveform that repeat among the background [21]. Instead of the absolute voltage of each sampling points, the EEG signal carries more information in its rhythm. Also, the intensity of signal received by localised electrodes on the scalp could provide information about the position of the brain regions that are activated [21]. By comparing sound generated by different channels with recording of close located electrodes, the relationship of coexist rhythm (represent by beats in sonified sound) represent the correlation between the recorded brain regions [21]. This is very useful in the detection of seizure as the intensity of beats would increase during seizure activity [21]. With detailed study of the characteristics of diseases, then combine with detecting and mapping methods, the translated audio would provide a well indication of certain body conditions. The success of the sonification in biofeedback will also support the detection of disease with temporal pathologic changes and learn the imperceptible signals of the body [21].

Similar to EEG, the translated sound signal from EMG recording could provide general perceptions of the muscle activation time and exertion level [6]. Given with the basic muscle information, more specialised study could be focused on any specific information extraction with certain group of participants [6]. The most common sound mapping parameters include pitch, volume, attack time (time taken for the pitch to reach its maximum intensity) and spatial location (in stereo field) were tested by Peres et al [6]. It was suggested that the detection of muscle extraction level with all four parameters mapped gave the best outcome whereas the perception of activation time would be disturbed by varying attack time. The spatialisation mapping to different muscles helped with the distinguish of each participant muscle [6]. By knowing which and when the muscle was generated, patients could be trained to gain a better control of their muscle. The prototype developed in the study was aiming for non-specific muscle EMG recording. As it only has one recording site (one sensor), the mapping discussed is inspiring if more electrodes or better audio player (for instance, headphone, use both right and left channel to achieve a stereo image) is applied for better motor learning in the future.

Adding on to the systematic approach of mapping, the task-based sonification could be arisen for various feedback. Pauletto and Hunt [25] have studied on identify known variables, age, and knee condition (with or without symptom of osteoarthritis). This was a backward study with known input parameter (age and condition), and the aim of the study was to find the sound features that could be used for classification. The sonification method included was amplitude modulation (AM), which mapped the signal from each sensor to a sine oscillator. Different sensor was assigned to different oscillators (in harmonic frequencies, for pleasant sound) [25]. The result shown that within the samples, the roughness (timbre) of sonified sound was correlated to age. The roughness also reduced in those symptomatic subjects (bigger experimental sample is needed for proof) [25]. In comparison, the sonification algorithm in the prototype was purely frequency modulation (FM) and used one channel of data. Additionally, in the testing stage of mapping (in Matlab), both sound amplitude and frequency was varied with extracted EMG parameters, with different correlations (logarithm and multiple) examined. The information extracted was encoded with the instantaneous frequency of sine wave (last in the length of one time window). It is worth to test on the listener's sensitivities with different mapping methods and find the best mapping. It was suggested that the detectability of FM is better than the AM solely, and combined modulations (AM and FM) would yield the best model [25].

Real-time auditory biofeedback could also use biomechanical data as it also contains information of human motions. For instance, the gait training could take ankle angle as input of auditory biofeedback device and generate a feedback loop to assist the user for gait correction [26]. A standard movement pattern was performed and recorded as reference. The two sides of headphone each played the reference track and the sound synthesised for real-time movement. The task of user is simply to match two sounds by adjusting the ankle-angle in walking. As there is no visual participation in this feedback, blind people could also use this device. The device shown an accurate match of angles after feedback [26]. The source like angles, torques and forces is less complicated to map than EMG signal and the understanding of information (whether the correction is needed) is simplified. However, the detailed neuromuscular data is not involved so precise feedback such as strengthen a certain muscle is difficult to achieve.

The possible application of the built prototype covers diagnostic application, athlete training, exercise injury prevention, prosthesis control and novel musical instrument. The clinician would easily miss the real-time information in visual display of EMG signals. The simultaneous audio display would provide an additional perception to the missing data and acquire a more accurate and detailed diagnosis. Correct muscle recruitment could be stimulated with this device and the exercise injury of athlete could be reduced and performance could be improved [6]. The audio from sonification can be modified towards a more musical approach, then the body signals could generate a music with mixture of nature and artificial interpretations. The muscle tension detected under stress, and all other emotions could be explored and interacted with the music and then response with the nervous system.

The compact size broadens the possible places and conditions to apply this prototype. Being able to record and sonify raw EMG signal in real-time provides a general start and leaves space for task-based adaptations. The future work for this study includes:

1. Work on the neatness of the prototype, rearrange the components for a more stable connection
2. Improve the audio circuit, for instance, add transistor to control the current, then the volume of audio
3. Continue the study on the sonification methods, extract more EMG parameters for mapping
4. Test on the biofeedback application, with experiment (like the wrist control experiment described above). Participants should be giving opportunity to experience the prototype and learn the sonified sound first, then test could be on determine the level of understanding the information carried in sound and test the performance with specific motion task

Conclusion

The portable prototype built in this research successfully played the translated sound signal in real-time. The parameter mapping algorithm was developed with the study of forearm EMG recordings. The Raw output from the sensor carries more valuable information than Envelope output and was used as input source. The noise level was reduced by a self-built processing circuit. The sonification of EMG signal was initially achieved with mapping the windowed MAV to frequency in the prototype. The future work of the hardware should be focus on the audio circuit as the outcome of the biofeedback is highly dependent on the sound synthesis and its interaction with nervous system (psychoacoustics). The software (algorithm running in the prototype) should be optimised to achieve a better interpretation of EMG signal and generate an informative audio (musical) output. Most importantly, future work need to include an investigation of the biofeedback possibilities of this prototype with participants.

Reference

1. Khokhar, Z.O., Xiao, Z.G. and Menon, C. (2010) ‘Surface EMG pattern recognition for real-time control of a wrist exoskeleton’, BioMedical Engineering OnLine, 9:41.

2. SENIAM. Available at: http://www.seniam.org/ (Accessed: 16 May 2022).

3. Hermann, T., Hunt, A. and Neuhoff, J.G. (eds.). (2011) ‘The Sonification Handbook’. Logos Verlag, Berlin, Germany.
4. Minciacchi, D. (2006) ‘Translation from Neurobiological Data to Music Parameters’, Annals of the New York Academy of Sciences, Volume 999, Issue 1, pp.282-301.
5. Loeb, E.G., Gans, C. (1986) ‘Electromyography for Experimentalists’. The University of Chicago Press, Chicago and London.
6. Peres, S.C., Verona, D., Nisar, T. and Ritche, P. (2016) ‘Towards a systematic approach to real-time sonification design for surface electromyography’, Displays, Volume 47, pp. 25-31
7. Meredith, T. (2018) ‘A Mathematical Model of the Neuromuscular Junction and Muscle Force Generation in the Pathological Condition Myasthenia Gravis’, p. 15.
8. Dosen, S. et al. (2015) ‘EMG Biofeedback for online predictive control of grasping force in a myoelectric prosthesis’, Journal of NeuroEngineering and Rehabilitaion, 12(1).
9. Budzynski, T.H. and Stoyva, J.M. (1969) ‘An instrument for producing deep muscle relaxation by means of analog information feedback’, Journal of Applied Behaviou Analysis, 2(4), pp. 231-237.
10. Hermann, T. (2008) ‘Taxonomy and Definitions for Sonification and Auditory Display’, International Conference on Auditory Display.
11. Sanyal, S., Nag, S., Banerjee, A. and Ghosh, D. ‘Music of brain and music on brain: a novel EEG sonification approach’, Cognitive Neurodynamics 13, pp. 13-31.
12. Väljamäe, A., et al. (2013). ‘A review of real-time EEG sonification research’, In: International Conference on Auditory Display 2013, pp85-93.
13. Dosen, S., Markovic, M., Somer, K. and Farina, D. (2015) ‘EMG Biofeedback for online predictive control of grasping force in a myoelectric prosthesis’. Journal of NeuroEngineering and Rehabilitation. 12, 55.
14. Grosshauser, T., et al. (2012) ‘Wearable Sensor-Based Real-Time Sonification of Motion and Foot Pressure in Dance Teaching and Training’, Journal of the Audio Engineering Society, 60(7), pp 580-589.
15. MyoWare Muscle Sensor (AT-04-001) DATASHEET, Available at: https://cdn.sparkfun.com/assets/learn_tutorials/4/9/1/MyoWareDatasheet.pdf (Accessed: 16 May 2022).

16. Chu, J.-U., Moon, I. and Mun, M.S. (2006) ‘A Real-Time EMG Pattern Recognition System Based on Linear-Nonlinear Feature Projection for a Multifunction Myoelectric Hand’, IEEE Transactions on Biomedical Engineering, 53(11), pp. 2232–2239.
17. Amrutha, N., Arul, V.H. (2017). ‘A Review on Noises in EMG Signal and its Removal’. International Journal of Scientific and Research Publication, Volume 7, Issue 5.
18. Raez, M.B.I, Hussain, M.S. and Mohd-Yasin, F. (2006) ‘Techniques of EMG signal analysis: detection, processing, classification and applications’, Biological Procedures Online,8, pp. 111-35

19. Bonato P, D'Alessio T, Knaflitz M. ‘A statistical method for the measurement of muscle activation intervals from surface myoelectric signal during gait’, IEEE Trans Biomed Eng. 1998 Mar, 45(3).

20. Davies MR, Reisman SS. Time frequency analysis of the electromyogram during fatigue. Proceedings of the 20th Annual Northeast Bioengineering Conference 1994; pp. 93-95.

21. Baler, B., Hermann, T. (2009) Sonification: listen to brain activity. In: Haas, R., Brandes, V. (eds) Music that works. Springer, Vienna.
22. Rickard, N.S. (2004) ‘Intense emotional responses to music: a test of the physiological arousal hypothesis’, Psychology of Music, 32(4), pp. 371–388.
23. Arduino Documentation, ‘Basic of PWM (Pulse Width Modulation), Available at: https://docs.arduino.cc/learn/microcontrollers/analog-output (Accessed: 16 May 2022).
24. Giggins, O.M., Persson, U.M. and Caulfield, B (2013) ‘Biofeedback in rehabilitation’, Journal of NeuroEngineering and Rehabilitation, 10.
25. Pauletto, S., Hunt, A. (2006) ‘The Sonification of EMG Data’, International Conference on Auditory Display.
26. Matsubara, M. et al. (2015) ‘WearableAuditory Biofeedback Device for Blind and Sighted Individuals’, IEEE MultiMedia, 22(1), pp. 68-73

Appendix
Arduino script:

1. Raw 1

```
void setup() {
  // put your setup code here, to run once:
Serial.begin(115200); // baud rate increased to 115200

}

void loop() {
  // put your main code here, to run repeatedly:
int sensorValue = analogRead(A1);
float voltage =sensorValue *(5.0 / 1024.0);
Serial.println(voltage);
}
```

2. Raw 2

```
const int analogInPin = A0;
int sensorValue = 0;
int mySensVals[500];

void setup() {
  // put your setup code here, to run once:
Serial.begin(9600);
}

void loop() {
  // put your main code here, to run repeatedly:
sensorValue = analogRead(analogInPin);
for (int i = 0; i < 500; i++) {
  sensorValue = analogRead(analogInPin);
  mySensVals[i]= sensorValue;

  delay(1);
}
for (int i = 0; i < 500; i++) {

 Serial.println(mySensVals[i]);
}
}
```

3. Arduino sonification

```
byte n = 50;
byte i=0; // data type
float Ref = 100;
int offset = 1024/2; // offset by 2.5V ... Correction
int RawArray[50];
long RawSum;
float WindowMav;
int  Gradient;
```

```
long GradientSum = 0;
float WindowGradient;
void setup() {
Serial.begin(9600);
  // put your setup code here, to run once

}

void loop() {
  // put your main code here, to run repeatedly:
//  read the analog input from A0

// int RawValue = analogRead(A0);
int EnvelopeValue = analogRead(A1);
// map the input range (digital reading from 0 to 1023)
RawArray[i] = abs(analogRead(A0) - offset);
 //Serial.println(analogRead(A0));


i = i+1;

if (i>=n) {
  //Serial.println("IF");
 for (int j = 0; j <= n-1; j++){
  RawSum += RawArray[j];
  WindowMav = RawSum/n;
 //Serial.println(j);
 }
 for (int k = 0; k <=n-2;k++){
  Gradient = RawArray[k+1] - RawArray[k];
  GradientSum += Gradient;
  WindowGradient = GradientSum/(n-1);
 }

 //for (int k = 0; k<=n-1; ++k){

 unsigned int Frequency= WindowMav*Ref;
 Serial.println(Frequency);
// noTone(9);
 tone (9, Frequency);
 i = 0;
 RawSum = 0;
 GradientSum = 0;
}

// play the pitch which follows the amplitude of the input reading
// in a time window

}
```

Matlab script:

```
%% plot EMG data
% Inputs
data = VarName2;
t = 6; % the total recording time (accounding to the timestamp)
%% map the data with time axis
f = numel(data)/t;
time = (0:1/f:(numel(data)-1)/f);
figure(1);
plot(time,data);
xlabel('time [s]');
ylabel('sample voltage [V]');

%% absolute value of the gradient
shifted_data = circshift(data,1);
gradient = (shifted_data - data);
abs_g = abs(gradient);

%% group each small time window (50ms) and parameters calculation within
% each window
n = t/0.05;   % number of time windows within the whole recording preiod
sample_size =floor(length(data)/n);
y0 = data(1:sample_size);
mav = zeros(1,n-1);
grad = zeros(1,n-1);

%% zeroed data
data_zero = data - mean(data);

%% For loop
% calculate the parameters within each time window
for i=1:n-1
    window = data_zero(i*sample_size:(i+1)*sample_size);
    w_mav = mean(abs(window));
    window_g = abs_g(i*sample_size:(i+1)*sample_size);
    w_grad = mean(window_g);
    w_rms = rms(window);
    mav(i) = w_mav;
    grad(i) = w_grad;

end
mav;
grad;
t_window = (0:0.05:(numel(mav)-1)*0.05); % time axis for parameters
figure(2);
plot(t_window,mav,t_window,grad)
legend('mean absolute value','gradient');
title('time windowed calculations');
xlabel('time [s]');
ylabel('parameter value');
%% Normalise the data
mav_norm = mav/(max(mav));
grad_norm = grad/(max(grad));

%% sine waves
fss =40000;
ts =(0:1/fss:0.05-(1/fss));
```

```
y(:,1) = sin(ts*(2*pi*log2(mav_norm(1))*400)); % first map MAV with the
frequency

for i = 2:numel(mav_norm)
    sineend = (y(end,i-1)); % starting to think about syncing to avoid cuts
    y(:,i) = sin(ts*(2*pi*log2(mav_norm(i))*400));
end
ys= y(:); % vectorise
figure(3);
plot(ys);
%% Amplification (amplitude mapping)
grad1 = repmat(grad_norm,numel(ts),1); % repeat copies of parameter array
in times of sine wave sampling point in one time window (40000 Hz in 0.05s,
so 200 copies)
grad2 = grad1(:) % vectorise
ys2 = grad2.*ys; % map wave amplitude with the mean absolute gradient value
figure(4);
plot(ys2); % the plot of audio data
title('mapped sound wave');



%% sound
audiowrite('EMG_sine_f_log3.wav',ys,fss); % sonified sound with only
frequency varying
audiowrite('EMG_sine_f&v_log3.wav',ys2,fss); % sound with both frequency
and amplitude varying
```